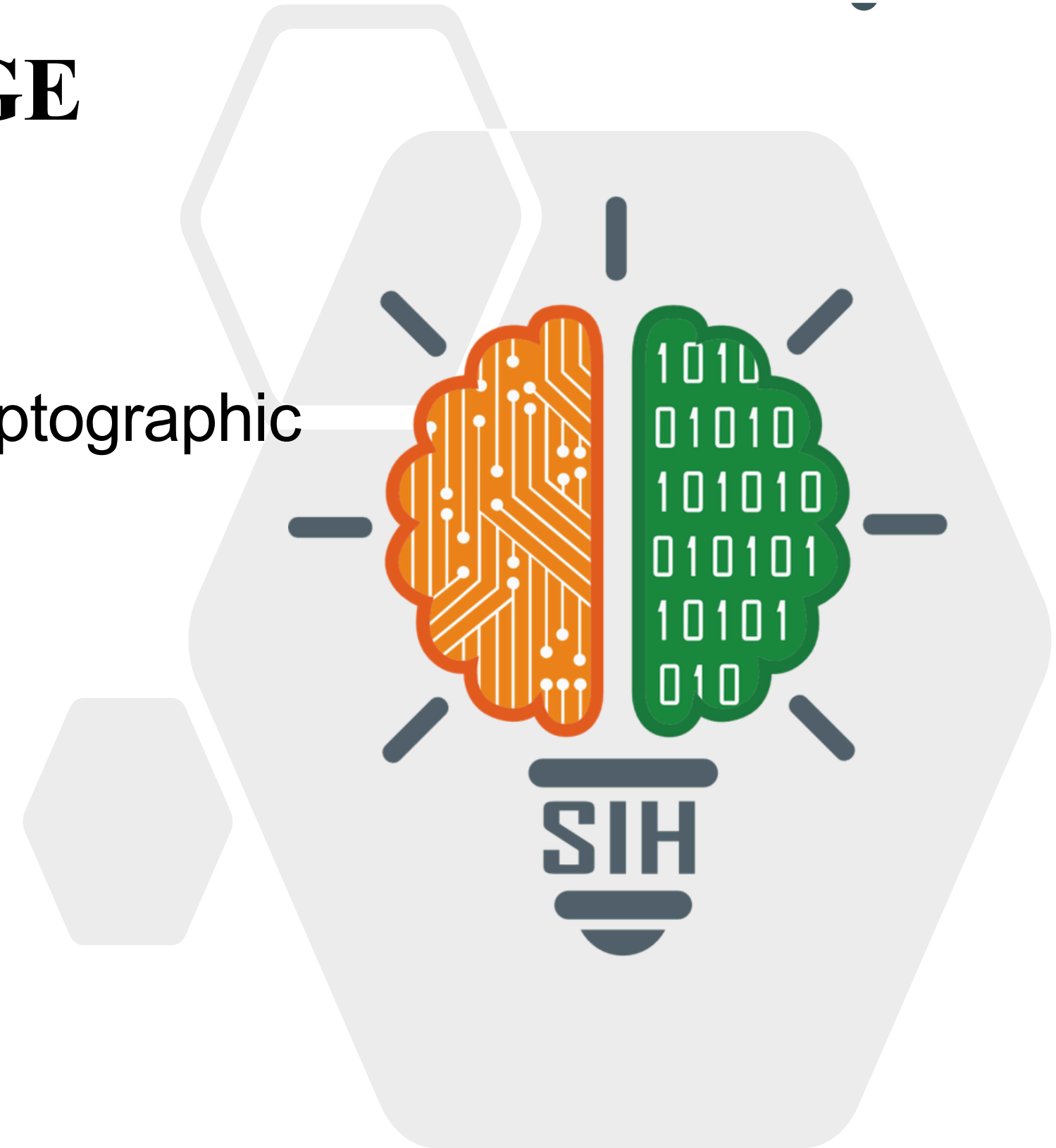


TITLE PAGE

- **Problem Statement ID** - SIH26164
- **Problem Statement Title** - Enterprise Cryptographic
Discovery & Analysis Tool (ECDAT)
- **Theme** - Blockchain & Cybersecurity
- **PS Category** - Software
- **Team ID** -
- **Team Name** - OHANA



CRYPTONEX — Enterprise Cryptographic Discovery & Analysis Platform

PROPOSED SOLUTION

CRYPTONEX is an enterprise cryptographic discovery platform that analyzes assets across **source code, dependencies, certificates, containers, binaries, and cloud environments**; assesses quantum risk using **Mosca's framework** and generates actionable **PQC migration** plans with standardized **CBOM output**.

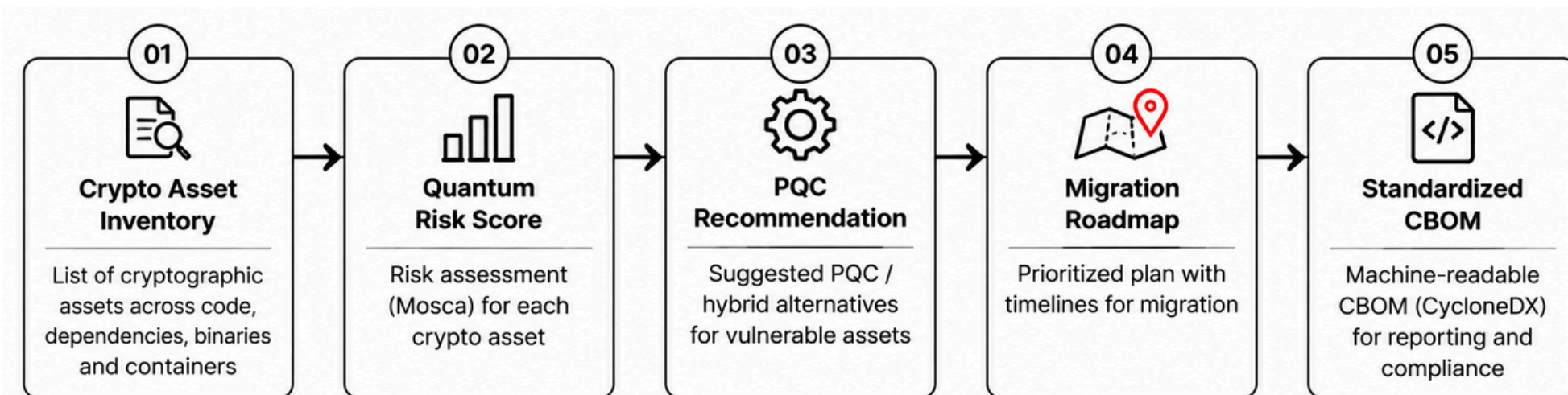
What we are solving

- Discovers cryptographic artefacts (algorithms, keys, certificates, libraries, containers, binaries)
- Assesses quantum risk using Mosca's algorithm and classifies by criticality
- Recommends PQC/hybrid alternatives based on risk profile
- Generates standardized CBOM + interactive GUI for visualization

CORE INNOVATIONS

- **Unified crypto scanning** : 6 local scanners + extensible hs/cloud collectors in 1 platform.
- **Interactive Mosca Lab**: Live $X+Y>Z$ risk modeling that most commercial tools skip.
- **Explainable assessment** : Every score split into Observed / Knowledge / Inferred / Assumed.
- **Actionable PQC Migration** : NIST-aligned fixes, sequenced into migration waves.

KEY OUTPUTS



TECH STACKS:



Backend
Core logic, analysis and orchestration



Scanning
Pattern matching for cryptographic artifacts



Secret Detection
Detects hardcoded keys, certificates and secrets



Dependency Analysis
Identifies cryptographic libraries and components

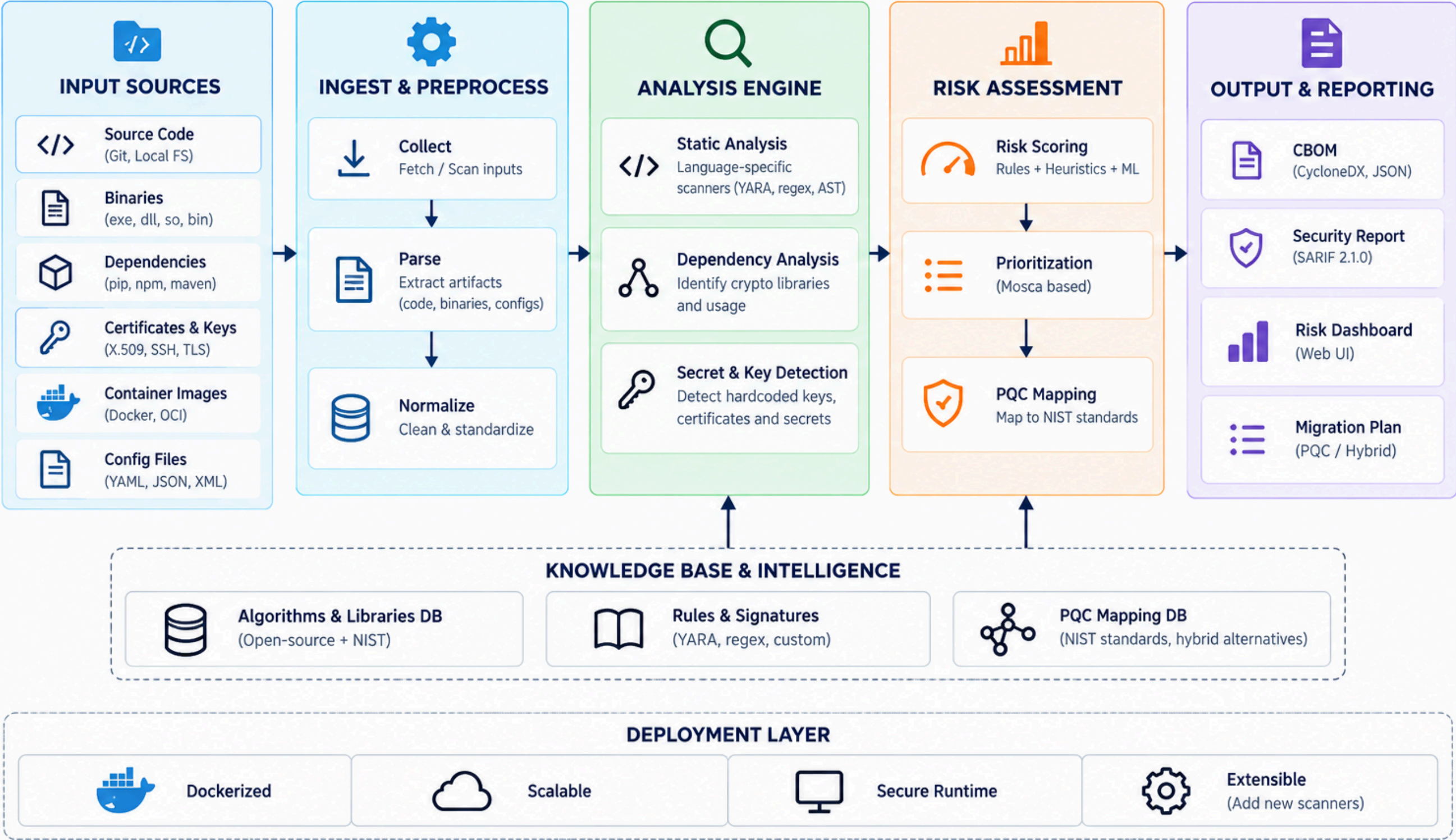


CBOM Generation
Generates standardized Cryptographic Bill of Materials



Deployment
Containerized and scalable execution

ARCHITECTURE:



Feasibility Analysis

- Modular scanner architecture supports **multiple crypto asset sources**
- Python-based implementation enables **rapid integration** of analysis libraries
- Supports **source code, dependencies, certificates, containers and binaries**
- Local processing enables sensitive enterprise **data to remain within the environment**
- Standardized **CBOM and SARIF** formats enable integration with existing security workflows

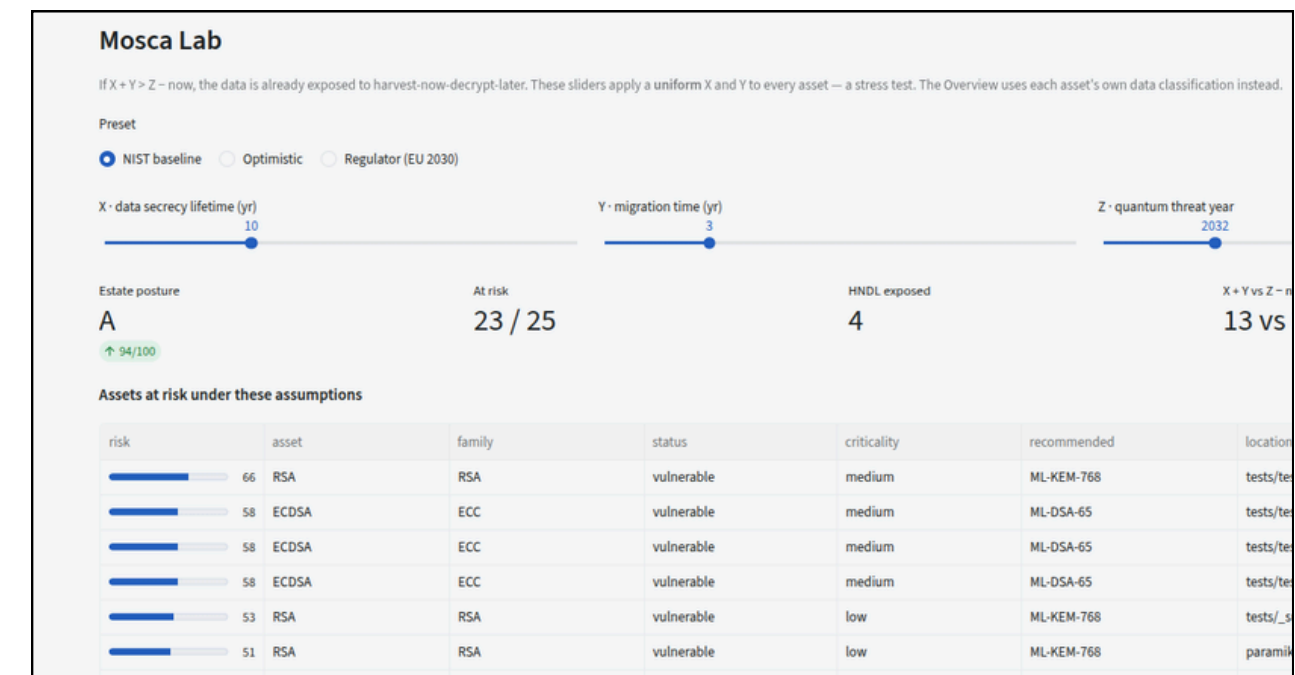
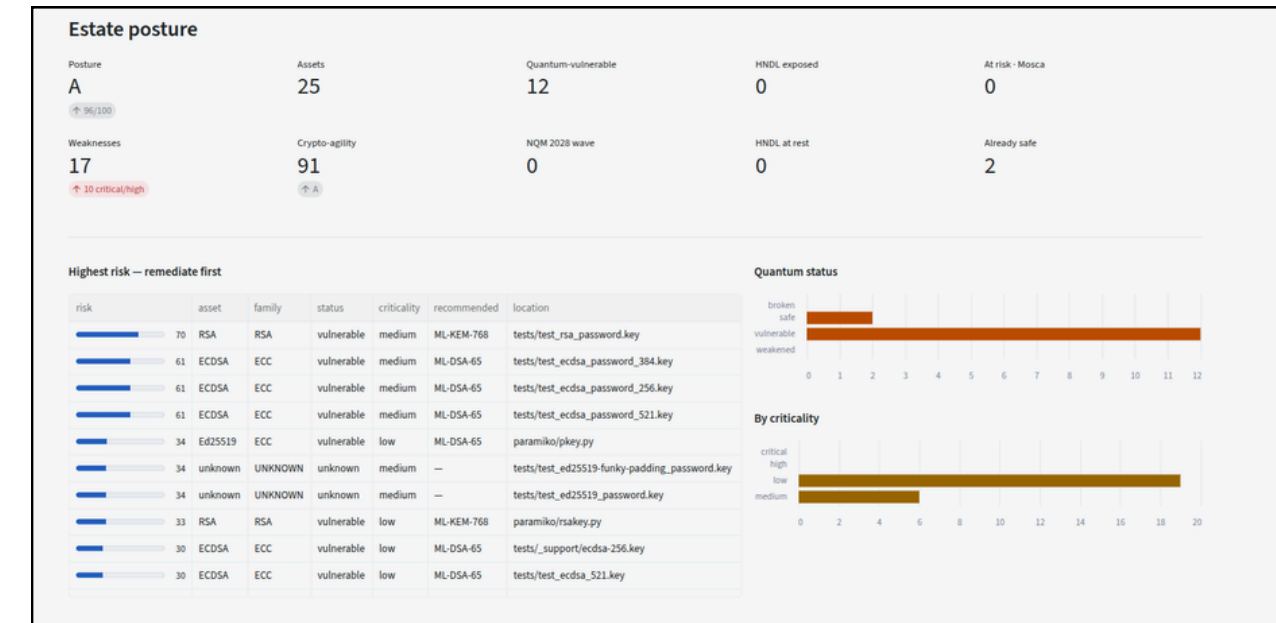
Risks:

- **False Positives**
Crypto usage may be incorrectly detected.
- **Hidden Cryptography**
Obfuscated or undocumented crypto may be missed.
- **Limited HSM / Cloud Coverage**
HSM and cloud discovery need real time integrations.
- **Quantum Assessment Uncertainty**
Mosca assumptions can vary by organization .

Overcome Strategy:

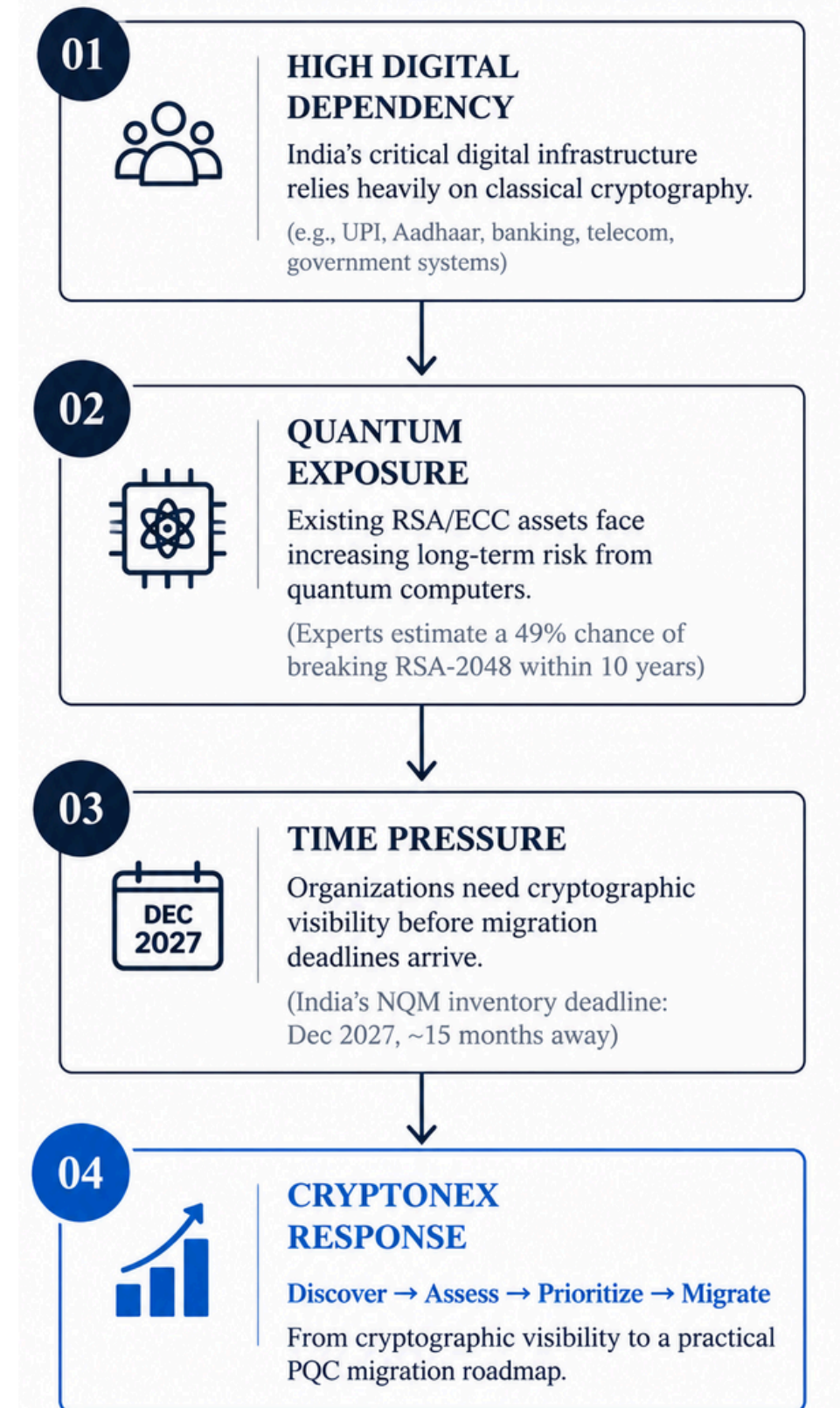
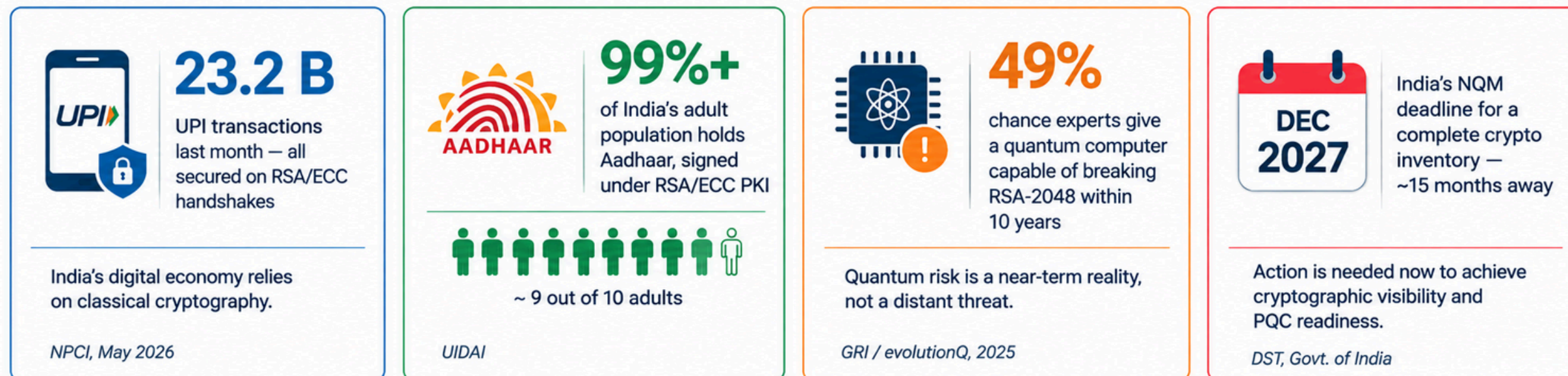
- **Evidence-Based Detection** : Confidence scores, code evidence, and repository benchmarking.
- **Extensible Design** : Modular scanners with defined coverage and HSM/cloud roadmap.
- **Configurable Risk Modeling** : Adjustable Mosca parameters with multiple assessment scenarios.

PROTOTYPE([demo video](#))



IMPACT AND BENEFITS

- **Social:** 23.2 billion UPI transactions a month and 99%+ of Indian adults on Aadhaar all rely on RSA/ECC today . CRYPTONEX finds that exposure before it's harvested and decrypted later.
- **Economic:** free vs. the \$100k+/yr enterprise tools it replaces; Mosca-based prioritization cuts migration cost by fixing the highest-risk assets first.
- **Strategic:** with experts now giving quantum a 49% chance of breaking RSA-2048 within 10 years, and India's NQM inventory deadline ~15 months out, CRYPTONEX gives defence, power, telecom and BFSI a ready, homegrown tool instead of starting from zero.



Github link: <https://github.com/nivas1899/cryptonex>

Prototype demo: <https://youtu.be/H-YUhDYYcWI?si=HhO-IAJxtFbLnqZs>

Research & References Sources

- NIST Migration to Post-Quantum Cryptography: <https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc>
- NIST Post-Quantum Cryptography Standards: <https://csrc.nist.gov/projects/post-quantum-cryptography>
- CycloneDX CBOM Specification: <https://cyclonedx.org/>
- NIST Transition to Post-Quantum Cryptography Standards: <https://csrc.nist.gov/pubs/ir/8547/ipd?>
- CISA, NIST & NSA — Post-Quantum Cryptography: <https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3498776/>